

Homework #1: Gradients of Sigmoid and Tanh

Proving the derivatives of two common deep learning activation functions.

1. Sigmoid Function Gradient

The sigmoid function is defined as:

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

Goal — show that:

$$\frac{d\sigma(x)}{dx} = \sigma(x)(1 - \sigma(x))$$

Proof

Rewrite $\sigma(x)$ as a power, with $u(x) = 1 + e^{-x}$ and $u'(x) = -e^{-x}$:

$$\sigma(x) = (1 + e^{-x})^{-1}$$

Applying the chain rule:

$$\frac{d\sigma}{dx} = -u(x)^{-2} \cdot u'(x) = \frac{e^{-x}}{(1 + e^{-x})^2}$$

Notice that:

$$1 - \sigma(x) = \frac{e^{-x}}{1 + e^{-x}}$$

Therefore:

$$\sigma(x)(1 - \sigma(x)) = \frac{1}{1 + e^{-x}} \cdot \frac{e^{-x}}{1 + e^{-x}} = \frac{e^{-x}}{(1 + e^{-x})^2}$$

Result:

$$\frac{d\sigma(x)}{dx} = \sigma(x)(1 - \sigma(x))$$

2. Tanh Function Gradient

The hyperbolic tangent is defined as:

$$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

Goal — show that:

$$\frac{d\tanh(x)}{dx} = 1 - \tanh^2(x)$$

Proof

Let $a(x) = e^x - e^{-x}$ and $b(x) = e^x + e^{-x}$. Note $a'(x) = b(x)$ and $b'(x) = a(x)$. Quotient rule:

$$\frac{d\tanh}{dx} = \frac{a'b - ab'}{b^2} = \frac{b^2 - a^2}{b^2}$$

Expanding the numerator:

$$b^2 - a^2 = (e^{2x} + 2 + e^{-2x}) - (e^{2x} - 2 + e^{-2x}) = 4$$

So:

$$\frac{d \tanh}{dx} = \frac{4}{(e^x + e^{-x})^2}$$

And we verify:

$$1 - \tanh^2(x) = \frac{(e^x + e^{-x})^2 - (e^x - e^{-x})^2}{(e^x + e^{-x})^2} = \frac{4}{(e^x + e^{-x})^2}$$

Result:

$$\frac{d \tanh(x)}{dx} = 1 - \tanh^2(x)$$

Summary

Function	Definition	Gradient
$\sigma(x)$	$1 / (1 + e^{-x})$	$\sigma(x)(1 - \sigma(x))$
$\tanh(x)$	$(e^x - e^{-x}) / (e^x + e^{-x})$	$1 - \tanh^2(x)$